JAWRA Journal of the American Water Resources Association

RESEARCH ARTICLE OPEN ACCESS

Leveraging High-Frequency Sensor Data and U.S. National Water Model Output to Forecast Turbidity in a Drinking Water Supply Basin

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Received: 6 May 2024 | **Revised:** 10 January 2025 | **Accepted:** 20 February 2025

Funding: This work was supported by the Cooperative Institute for Research to Operations in Hydrology (CIROH) through a National Oceanic and Atmospheric Administration (NOAA) Cooperative Agreement with the University of Alabama, NA22NWS4320003.

ABSTRACT

As high-frequency sensor networks increasingly enhance data-driven models of water quality, process-based models like the U.S. National Water Model (NWM) are generating accessible forecasts of streamflow at increasingly dense scales. There is now an opportunity to combine these products to construct actionable water quality forecasts. To that end, we couple streamflow forecasts from the NWM to a gradient-boosted decision tree algorithm (LightGBM) trained on 5+ years of high-frequency monitoring data to forecast in-stream turbidity levels in the Catskill Mountains, NY, USA. Results indicate LightGBM models are capable of relatively skillful predictions, which enable robust forecasts for 1–3 days lead times. LightGBM models offer improvements over a simplified linear model across the entire forecast horizon, and more spatially complex models are more resilient to error at shorter lead times (1–3 days). Moreover, interpretation of model features emphasizes high flows as a driver of turbidity in the region. Results suggest that interpretable, flexible, and efficient machine learning algorithms can produce capable water quality forecasts from streamflow forecasts and expand understanding of process dynamics. The use case illustrated here—to our knowledge the first NWM-based water quality forecast—underscores the potential to employ the NWM to expand national water quality forecasting capacity and can overall serve as a guide for similar efforts in basins across the country.

1 | Introduction

The implications of high turbidity and sediment concentration in riverine and other water bodies are manifold, including elevated transport of sorbed contaminants (e.g., Kronvang et al. 1997; Vidon et al. 2018; Underwood et al. 2017), impaired quality of drinking water resources (e.g., Wang, Gelda, et al. 2021; Mukundan et al. 2018; Mukundan, Pierson, Schneiderman, et al. 2013; Mukundan, Pierson, Wang, et al. 2013; Gelda and Effler 2007), and general degradation of aquatic ecosystems (e.g., Bilotta and Brazier 2008; Kaller and Hartman 2004; Kemp et al. 2011). Enhanced ability to forecast turbidity and sediment loading from a given flow event or range of hydrological conditions thus has substantial utility for the active management of aquatic systems, especially in the case of drinking water supply networks (e.g., Gelda and Effler 2007; Stevenson and Bravo 2019; Wang, Gelda, et al. 2021).

Prediction of turbidity (and closely associated suspended sediment concentration) is challenging in part due to the complex web

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Summary

- Post-processing National Water Model streamflow forecasts with machine learning models trained to predict water quality from observational data can provide capable turbidity forecasts and is an intriguing path forward for expanding national water quality forecasting capabilities.

of processes and feedbacks that drive concentration and loading during a given streamflow event. Concentration-discharge (cQ) relationships have traditionally been used to build a sediment rating curve that predicts concentration (or flux) as a function of discharge in the form of a power law (Asselman 2000; Gao 2008) (Equation 1)

$$C = \mathrm{aQ}^{\mathrm{b}} \tag{1}$$

(where C is concentration, Q is discharge, and a and b are empirically derived coefficients), but various additional factors complicate this relatively straightforward relationship and lead to considerable scatter around derived curves (Gao 2008; Mohanty et al. 2023; Vercruysse et al. 2017; Wymore et al. 2023). Complications and departures from the general cQ relationship for a given discharge (i.e., hysteresis) may have myriad drivers, including antecedent moisture conditions (e.g., Hamshaw et al. 2018; Onderka et al. 2012; Seeger et al. 2004), rainfall intensity (e.g., De Girolamo et al. 2015; Nadal-Romero et al. 2008; Zabaleta et al. 2007), spatial organization of potential sources (e.g., Buendia et al. 2016; Gellis 2013; Lawler et al. 2006), and network connectivity (e.g., Hamshaw et al. 2018; Keesstra et al. 2019; Wang and Steinschneider 2022), as well as interactions therein.

To better encompass the hysteresis effects that complicate the general cQ relationship and preclude straightforward prediction, additional approaches can be employed beyond a simple direct relation. One often applied technique is quantile regression, which estimates unique regression equations for each quantile of the observed response variable and thus generates a distribution of potential concentrations for each value of discharge (Cade and Noon 2003; Mukundan et al. 2018; Shiau and Chen 2015). This can be conceptualized as an extension of the linear cQ approach that incorporates probabilistic principles and is capable of improving predicative ability by more fully capturing the variability present in collected data without considering additional explanatory variables (Mukundan et al. 2018). Another approach is the construction of higher-order models of concentration that incorporate predictors beyond at-a-point discharge magnitude (i.e., Q), such as those drivers mentioned above (e.g., antecedent moisture conditions, rainfall intensity, etc.) (Hamshaw et al. 2018; Onderka et al. 2012; Seeger et al. 2004).

One avenue toward building enhanced data-driven models of water quality is the collection of high-frequency hydrologic and water quality data using field-deployed sensors. These sensors (and sensor networks) (e.g., Hamshaw et al. 2018; Kemper et al. 2019; Wang, Davis et al. 2021) have proliferated over the last decade-plus and enable the collection of hydrobiogeochemical data at the robust spatial (e.g., nested watersheds) and temporal (e.g., 15-min, hourly) scales necessary to expand beyond relatively simple cQ relations (Hart and Martinez 2006; Kirchner et al. 2004; Rode et al. 2016). When this type of data is paired with machine learning/artificial intelligence (ML/AI) approaches—unbound by the assumptions of linearity inherent in more traditional techniques and capable of parsing large quantities of data—improved estimation of concentrations and fluxes is possible (Zhong et al. 2021).

In addition to refining predictions of concentration, enhanced models can also be used to forecast turbidity and sediment (or parameters such as temperature, phosphorus, dissolved oxygen, or any other constituent of concern in a given watershed for which abundant monitoring data exist) when paired with forecasts of hydrometeorological variables. Though traditional cQ-based models could be used for this undertaking, machine learning water quality models built on hydrological drivers (e.g., Hamshaw et al. 2018; Javed et al. 2021) may have enhanced potential to produce skillful forecasts of water quality parameters from hydrometeorological forecasts (e.g., streamflow). This is especially true of late: as monitoring networks continue to enrich our ability to predict water quality, high-resolution streamflow forecasts such as those offered by the U.S. National Water Model (hereafter NWM) (Cosgrove et al. 2024) have become increasingly available and accessible, and there is thus a distinct opportunity to leverage these capabilities to produce robust water quality forecasts at actionable spatial and temporal scales (e.g., Wade et al. 2024).

The NWM is a large-scale hydrologic forecasting model developed and maintained by the U.S. National Oceanic and Atmospheric Administration (NOAA) that is based on the Weather Research and Forecast-Hydrologic model (WRF-Hydro) (Cosgrove et al. 2024; Johnson et al. 2023; Read et al. 2023). Executed on a 1-km resolution gridded land surface and using a channel routing network derived from the National Hydrography Dataset Plus Version 2 (NHDPlus), version 2.1 (v2.1) of the NWM provides streamflow forecasts at 2.7 million reaches across the continental United States, Hawaii, and Puerto Rico (Cosgrove et al. 2024). Several different model configurations provide forecasts with a variety of look-ahead times, including a Short-Range configuration with an 18-h look-ahead window, a multi-member Medium-Range configuration with an 8.5–10-day window, and a Long-Term configuration with a 30-day window (Cosgrove et al. 2024). In short, the NWM can provide temporally high-resolution forecasts at a relatively dense spatial resolution, making it an ideal partner for data-driven models of water quality built on spatially and temporally dense sensor networks. However, to our knowledge, this coupled approach is an opportunity that has not yet been attempted.

In this paper, we combine data-driven models of turbidity with streamflow forecasts from the NWM, detailed above, and from the Northeast River Forecast Center (NERFC) (see Section 2.3), to build forecasts of turbidity in a critical drinking water supply watershed where reservoir operations are frequently impacted by elevated turbidity. Flow forecasts from multiple sources are used to demonstrate the generalizability of the approach and illustrate how it may be applied using emerging tools, like the NWM, or established operational approaches, like the official National Weather Service streamflow forecasts generated by NERFC.

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Turbidity forecasts are produced by feeding NWM and NERFC streamflow forecasts to a quantile regression model (Mukundan et al. 2018) and a ML/AI model—the LightGBM implementation of gradient-boosted decision trees (Ke et al. 2017)—and are then evaluated against benchmarks generated by feeding observed flow to those same models. In the subsequent sections, we first present our modeling workflow, complete with discussion and justification of model choices, feature selection, and forecasting approach. We then present model performance results, both in a benchmark and forecasting implementation, using a suite of commonly employed evaluation metrics. Finally, we consider performance patterns and how a given model may improve prediction in a particular situation, discuss how the approach delineated here might broadly illustrate the abilities of the NWM to act as a water quality forecasting tool, and examine the process-based insights generated by our models.

Overall, we aim to demonstrate the capabilities of data-driven models to generate skillful water quality (turbidity) forecasts from hydrometeorological forecast model outputs and illustrate the value added when a flexible, nonlinear ML/AI approach (LightGBM) is used in addition to a simplified, more commonly employed extension of a linear approach (here, quantile regression). We hypothesize that (1) the ML/AI method can better capture the non-linear relationship between turbidity and discharge, improving our ability to produce turbidity forecasts from forecasts of streamflow, and (2) this capability is further enhanced by the incorporation of the network-scale streamflow forecasts offered by the NWM. As a use case, we demonstrate our approach in a critical drinking water supply watershed in the Catskill Mountains of southeastern New York, USA. Study findings have the potential to improve the anticipation of turbidity loading within the test catchment, which drains to a major New York City water supply reservoir with mandated levels of turbidity non-exceedance (Gelda et al. 2009). Given the widespread availability of NWM forecasts, results will also help guide efforts to forecast, anticipate, and manage turbidity in watersheds throughout the country and, more generally, inform similar approaches in basins worldwide.

2 | Study Site & Data Availability

2.1 | Upper Esopus Creek Watershed

The upper Esopus Creek watershed (Figure 1) is a ~497 km^2 heavily forested, mountainous catchment in the Catskill Mountains of southeastern New York State (Davis et al. 2009; Mukundan et al. 2018; Wang, Gelda, et al. 2021). Relief in the basin is relatively pronounced—maximum elevation in the watershed is 1274 m above sea level at the summit of Slide Mountain, descending to 178 m by the watershed outlet—and the various streams of the catchment are consequently high-energy, high-gradient channels (McHale and Siemion 2014). Bedrock consists of repeating sequences of Devonian sandstones, mudstones, and conglomerates (Ver Straeten 2013), and surficial deposits are a complex combination of Pleistocene glacial and proglacial sediments alternatively covered or replaced by Holocene alluvium and colluvium (Davis et al. 2009; Rich 1934). Exposure of and erosion into glacial legacy sediments along valley bottom stream margins is a primary source of suspended sediment and turbidity within the basin, especially during high-flow events, and streambank erosion overall dominates fluvial sediment sources (Mukundan et al. 2018, 2015; Mukundan, Pierson, Schneiderman, et al. 2013). Glacial sedimentary deposits accessed and activated during high-flow events can result in greatly elevated turbidity and generate suspended sediment concentrations (SSC) of substantial magnitude (~10^3 mg/L) (McHale and Siemion 2014). Activated deposits can persist as a source of turbidity and SSC for some time after an event, resulting in relatively elevated sediment and turbidity concentrations during subsequent flows (Ahn et al. 2017; Siemion et al. 2023; Wang, Gelda, et al. 2021; Wang, Davis et al. 2021).

The watershed drains to Ashokan Reservoir (Figure 1), the second largest reservoir by volume within the New York City Water Supply System (NYCWSS), which supplies drinking water to approximately 10 million people in New York City and upstate counties. NYCWSS operates under a filtration avoidance determination granted in conjunction by the US Environmental Protection Agency and the New York State Department of Health (Mukundan et al. 2015). Under federal regulatory standards, turbidity levels within the NYCWSS must not exceed 5 NTUs in the terminal reservoir, and large-magnitude runoff events that result in elevated turbidity within the Ashokan can limit the usage and supply of drinking water from the basin (Effler et al. 1998; Gelda et al. 2009; Mukundan, Pierson, Wang, et al. 2013). Management of turbidity and suspended sediment sources is thus of substantial importance for water resources (NASEM 2018, 2020; Siemion et al. 2023; Siemion et al. 2016; Wang, Davis et al. 2021), and the watershed is accordingly densely monitored for discharge, turbidity, and SSC (McHale and Siemion 2014; Siemion et al. 2021).

2.2 | Observational Data

In the study here, we concentrate on turbidity and discharge measurements from 10 primary monitoring stations that represent the output of important tributary watersheds as well as that of the total upper Esopus Creek catchment (Table 1; Figure 1). Parameters are measured at 15-min intervals at all stations. SSC estimations for each station are derived from turbidity measurements using a rating curve approach (Siemion et al. 2021). We focus on turbidity as the primary data source due to the relatively greater temporal length of turbidity records compared to SSC, the usage of turbidity as the operating parameter for the NYCWSS (Mukundan et al. 2015; Mukundan, Pierson, Schneiderman, et al. 2013), and the relatively strong correlation between turbidity and SSC in the Esopus Creek watershed (McHale and Siemion 2014; Siemion et al. 2021; Wang, Davis et al. 2021). Turbidity and discharge records coincident across all these 10 stations extend from December 15, 2016 to the present day (2024), and we concentrate our analysis on the period between the start of the synchronous record and September 20, 2023 (roughly corresponding to water years 2017–2023).

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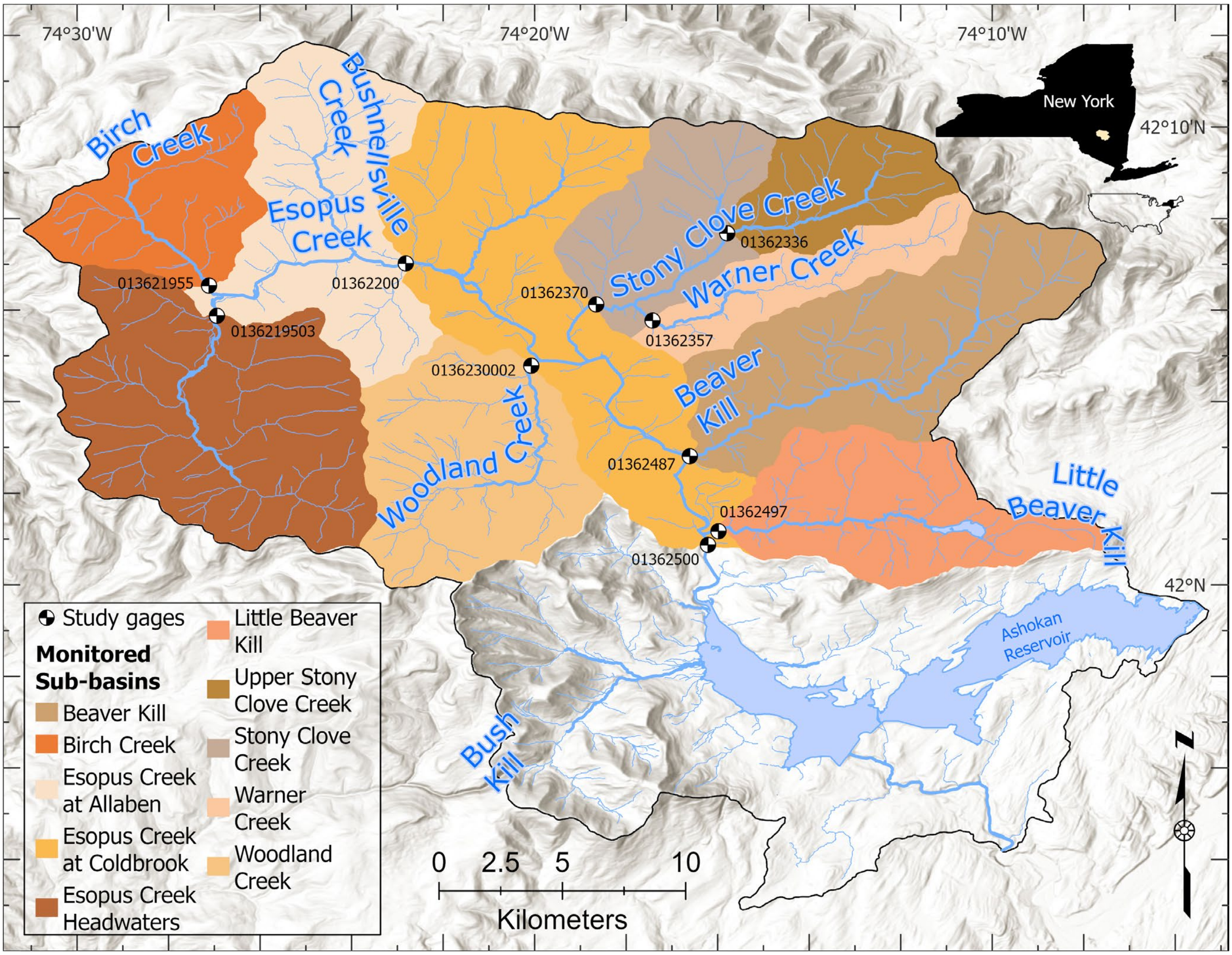

FIGURE 1 | The Upper Esopus Creek watershed with sub-basins denoted by color. Dichromatic dots are U.S. Geological Survey gages used in this study; numbers represent the USGS site ID number associated with each gage. Black dots are monitoring gages not considered here. The colored portion of the watershed drains to the Esopus at Coldbrook (01362500) USGS gage; the black outline denotes the full watershed of the Ashokan Reservoir.

2.3 | Forecast Streamflow Data

NWM streamflow forecasts considered here are those generated by NWM v2.1 (Cosgrove et al. 2024), which was the operational version of the NWM from April 21, 2021 to September 20, 2023. Specifically, we concentrate on the operational outputs of the seven-member Medium-Range (MR) configuration, which is an ensemble forecast with a 10- and 8.5-day look-ahead window (members 1 and 2–7, respectively) (Cosgrove et al. 2024). Forecasts for each of the seven members are executed at 6-h intervals beginning at midnight Coordinated Universal Time (i.e., 00Z) each day. Each member is forced by Global Forecast System (GFS)—a U.S. National Weather Service gridded weather forecast model that projects dozens of variables, including temperature, precipitation, soil moisture, etc.—outputs, with member 1 forced by the most recent GFS output and members 2–7 by increasingly older GFS cycles. Notably, each member starts with the same set of initial conditions, which include "nudged" discharge values generated by data assimilation of observed flow at USGS gages (Cosgrove et al. 2024). NWM discharge forecasts are made for 2.7 million reaches within the continental United States at hourly timesteps; downloaded forecasts for this study are for those reaches corresponding to the primary gage locations within the Esopus Creek watershed (Table 1).

River stage forecasts from the NERFC are generated using a hydrologic prediction model built on the Sacramento Soil Moisture Accounting Model (Burnash et al. 1973) that is forced by Quantitative Precipitation Forecasts, which themselves are produced via a complex blend of numerical model output, observational data, and statistical methods (Adams 2016; National Weather Service (NWS) 1999). NERFC forecasts are generally issued daily around 11 am local time, unless a major runoff event is underway, at which point they are output with greater frequency (i.e., sub-daily). Stage forecasts have a 72-h look-ahead window and are output at 6-h timesteps. NERFC forecasts are generated for ~236 locations in the northeastern United States, only one of which falls within the Esopus Creek catchment and corresponds to the lowermost (i.e., outlet) Esopus Creek at Coldbrook gage (Table 1). NERFC forecasts (and those from the other 12 RFCs located around the country) are the official flood forecasts of the U.S. Government (Adams 2016).

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TABLE 1 | Esopus Creek sub-basin characteristics.

USGS site ID	USGS station name	DA (km^2)	Q record start	T_n record start	COMID	NWM nudge	Stream slope (m/m)
0136219503	Esopus Creek Below Lost Clove Rd. at Big Indian NY	76.7	10/12/2016	9/1/2016	6189668	Yes	0.027
013621955	Birch Creek at Big Indian NY	32.4	9/30/1998	10/1/2016	6189634	Yes	0.035
01362200	Esopus Creek at Allaben NY	165	10/1/1963	12/12/2016	6189624	Yes	0.016
0136230002	Woodland Creek Above Mouth at Phoenicia NY	53.4	8/1/2003	11/10/2016	6189648	Yes	0.031
01362336	Stony Clove Cr at Janssen Rd. at Lanesville NY	24.0	11/15/2016	11/3/2016	6189746	No	0.074
01362357	Warner Creek near Chichester NY	22.6	5/29/2012	5/30/2012	6189748	Yes	0.035
01362370	Stony Clove Creek Blw Ox Clove at Chichester NY	80.0	2/1/1997	11/23/2010	6189644	Yes	0.026
01362487	Beaver Kill at Mount Tremper NY	64.7	9/30/2010	11/17/2010	6189750	Yes	0.020
01362497	Little Beaver Kill at Beechford near Mt. Tremper NY	42.7	10/1/1997	11/16/2010	6189752	Yes	0.005
01362500	Esopus Creek at Coldbrook NY	497	10/1/1931	10/1/2016	6189754	Yes	0.009

Note: Q, discharge; T_n, turbidity; COMID, reach ID in the National Hydrography Dataset (NHD). Note that all gages are currently operational as of this writing (2024). USGS, United States Geological Survey. National Water Model Nudge indicates whether (or not) gage discharge data is assimilated by the model.
Abbreviation: DA, drainage area.

3 | Methods

3.1 | Data Download & Pre-Processing

3.1.1 | Observational Data

Instantaneous (15-min), non-provisional discharge and turbidity data were downloaded for the period October 1, 2015–September 30, 2023 at each gage location (Table 1) via the USGS National Water Information System (U.S. Geological Survey, 2024a). Turbidity data for the Coldbrook station were also retrieved from a USGS data release (Siemion 2024). Note that the download period is larger than the period of analysis to enable the calculation of antecedent conditions over the entire study period. Data downloads were done using the dataRetrieval package (DeCicco et al. 2023) in the R statistical computing software (R Core Team 2023).

Turbidity data across the study period were collected by two different turbidity probes (a Forest Technology Systems DTS-12 and an Analite Observator NEP-5000), the latter of which replaced the former on June 30, 2022 at Coldbrook. To make turbidity readings transferable between sensors and build a continuous time series of equivalent readings, data from the Analite were transformed using a linear regression built against contemporaneously measured DTS-12 values at several sites (Table 1, Figure S1) (Messner et al. 2023).

To harmonize observational data with hourly streamflow forecast outputs, 15-min data were aggregated to an hourly time step. For discharge, a simple arithmetic mean was used; for turbidity, a flow-weighted hourly mean concentration was calculated by multiplying instantaneous turbidity by the corresponding discharge value, integrating over the hour, and dividing by the total water load (i.e., discharge integrated over the hour). Data gaps of 24-h or less for both turbidity and discharge were infilled using a simple linear interpolation; data from larger gaps were dropped.

3.1.2 | Forecast Data

NWM operational medium-range forecast data were downloaded for the period October 1, 2022 to September 20, 2023. This time period was chosen to optimize the split between observational training data and forecast testing. Only a single forecast execution (00z) was downloaded for each day, and the mean of the seven-member forecast at each timestep was taken for each daily execution to produce a single-value time series of streamflow. Data downloads were performed with the assistance of the nwmTools package (Johnson 2023).

NERFC forecast data for the same period were downloaded via the Iowa Environmental Mesonet bulk download function. Only a single NERFC forecast—the forecast issued closest to 11 am local time—was downloaded for each day. NERFC

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issues water-stage forecasts, so forecasted stage values were back-transformed to discharge using the stage-discharge rating curve for Esopus Creek at Coldbrook (accessed via the USGS WaterWatch Toolkit [U.S. Geological Survey, 2024b]). Because the forecasting model used by NERFC originally outputs discharge that is then internally transformed to stage before communication to the public, this back-transformation likely does not result in any loss of information.

3.2 | Overall Modeling & Forecasting Approach

In total, we developed three data-driven regression models of turbidity: a quantile regression quadratic model (Mukundan et al. 2018) and two LightGBM models (Ke et al. 2017) (Figure 2). Each model incorporates different combinations of predictors (see Sections 3.3 and 3.4) that generate estimates of turbidity from discharge and various discharge-derived parameters. Once built, each model was fed different discharge forecasts (see Section 3.4). The quantile regression model was fed both NWM and NERFC forecasts, as was one of the LightGBM models. The second LightGBM model, built using spatially distributed data, was only fed NWM forecasts because NERFC forecasts are not available for locations other than the Coldbrook gage (Figure 2). In total, this approach produced five different turbidity forecasts: two generated using NERFC outputs that have a forecast horizon of 1–3 days, and three generated from NWM output with a forecast horizon of 1–8 days (Figure 2). The sections that follow provide further details on each step of the approach. LightGBM was chosen over other available ML/AI algorithms because it is a powerful, interpretable technique optimized for efficient learning with large-scale datasets that contain many features. To achieve particular efficiency, LightGBM utilizes Gradient-Based One-Side Sampling to prioritize data instances with large gradients, reducing the size of the data considered, and uses exclusive feature bundling to bin mutually exclusive features, decreasing the total number of features (Ke et al. 2017). Additionally, gradient-boosted decision tree models—of which LightGBM is one—are more readily interpretable than more complex approaches such as neural networks, have been shown to frequently achieve roughly equivalent performance, and may be better suited for data with skewed distributions or other types of irregularities (e.g., data gaps) (McElfresh et al. 2023). Given the goal of this study is to build a capable forecasting approach that also enhances process understanding in the basin, and considering the dataset here is large, skewed, and contains a few unavoidable irregularities (i.e., missing data during station outages), LightGBM was deemed an appropriate choice of algorithm.

3.3 | Model Descriptions

Because of their different capabilities, each model incorporated the predictor variable of discharge in a different manner. The

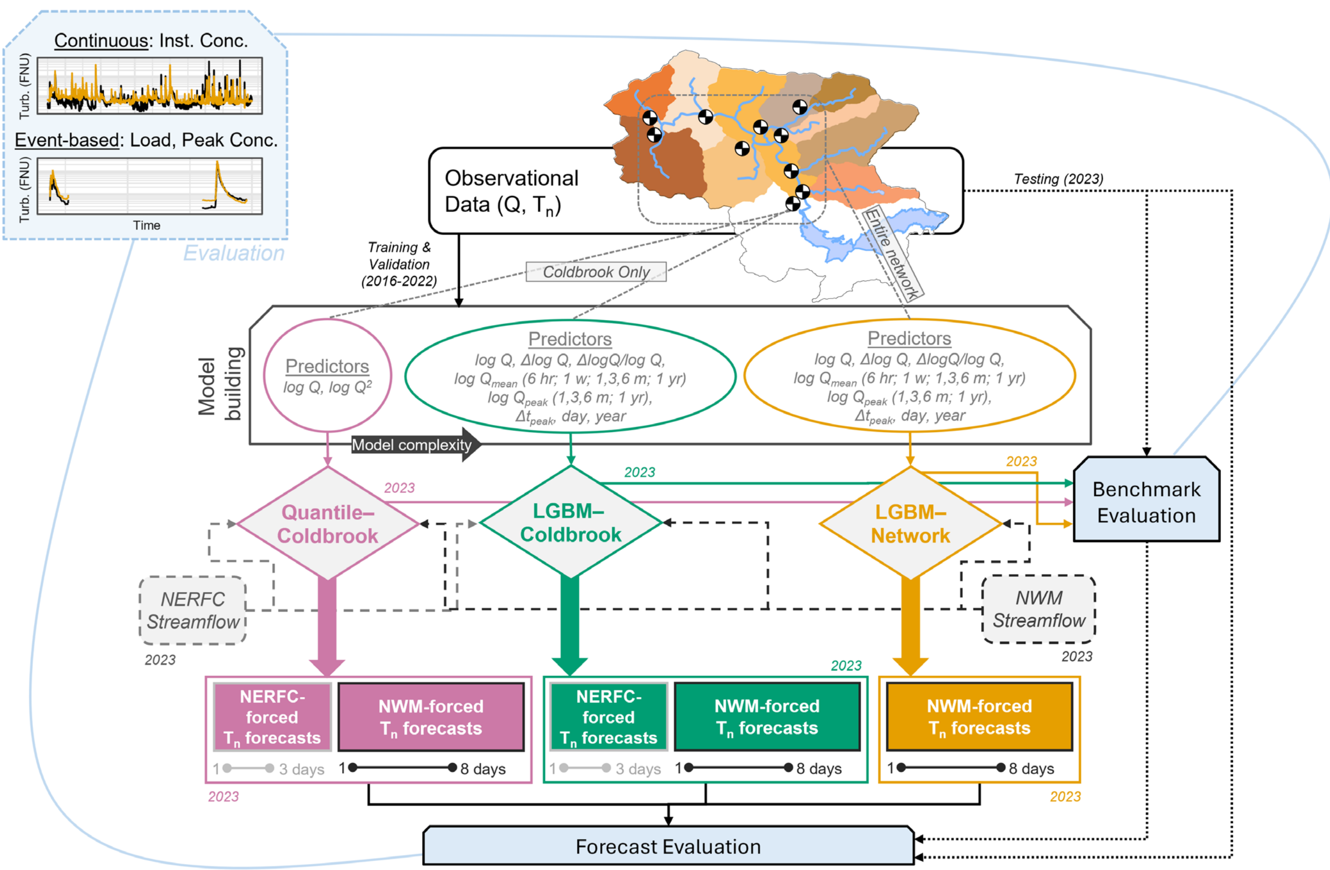

FIGURE 2 | Model building, forecasting, and evaluation workflow for leveraging streamflow forecasts to predict water quality. Dates correspond to the water year(s) of input (or, in the case of T_n forecasts, output) for each step. Q, discharge; T_n, turbidity; NERFC, Northeast River Forecast Center; NWM, National Water Model; LGBM, LightGBM. "Conc" is short for concentration, and "inst" is short for instantaneous.

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quantile regression model (hereafter Quantile-Coldbrook) used at-a-point observations of discharge and discharge squared (Equation 2); each LightGBM model used various combinations of observed discharge and antecedent conditions. The first LightGBM model (hereafter LGBM-Coldbrook) considered discharge and antecedent discharge conditions as measured only at the outlet Esopus at Coldbrook, NY gage, while the second (hereafter LGBM-Network) considered these parameters both at Coldbrook and as measured at the other study gages in the watershed (Table 1, Figure 1).

The Quantile-Coldbrook model regresses observed log-transformed turbidity (T_n) against observed log-transformed discharge (log Q) and log-transformed discharge squared (log $Q)^2$ at different turbidity quantiles (q) following Equation (2).

$$\log\left(T_{n_q}\right) = \beta_{0,q} + \beta_{1,q} \log(Q) + \beta_{2,q} \left[\log(Q)\right]^2 \quad (2)$$

Regression coefficients are determined by minimizing the sum of asymmetrically weighted absolute error (i.e., for large q, more weight is given towards minimizing underprediction, and vice versa for small q) (Mukundan et al. 2018; Wang, Gelda, et al. 2021). This approach yields different regression coefficients for each turbidity quantile and overall generates a probabilistic distribution of turbidity values for a given discharge. For comparison purposes, we focus here on the model generated by regression against the median ($q = 0.5$). Though this does not leverage the probabilistic capabilities of the quantile approach, it retains some benefits over traditional linear regression simply because it does not assume that data are normally distributed about a mean (Mukundan et al. 2018; Wang, Gelda, et al. 2021). No additional predictors were added to the quantile regression equation because the model in this general form is currently in operational use for turbidity prediction by the New York City Department of Environmental Protection (NYCDEP) and we wished to examine the performance of this straightforward, relatively simple specific model rather than generally compare the performance of (quasi)linear models to ML/AI techniques.

Both LightGBM models use several additional predictors beyond discharge (log Q). Informed by prior work on drivers of turbidity and suspended sediment in Esopus Creek (Ahn et al. 2017; Ahn and Steinschneider 2018; Mukundan, Pierson, Schneiderman, et al. 2013; NYCDEP 2023; Wang, Gelda, et al. 2021; Wang, Davis et al. 2021), we incorporated various other predictors derived from the discharge record (Table S1, Figure 2): the difference between log Q at time t and log Q at the previous timestep (Δlog Q); that difference normalized by the log of discharge at time t (Δlog Q/log Q); mean log-transformed discharge (log Q_{mean}) over a prior period of several different lengths (6-h, day, week, 1-, 3- and 6-month, and year); maximum log-transformed discharge (log Q_{peak}) over a prior period of different lengths (month, 3-month, 6-month, and year); the time between each peak and time t (Δt_{peak}); the Julian day (day); and water year (year). Respectively, for time t, these hydrological variables represent (or serve as a proxy for) the limb of the hydrograph (rising, falling); the intensity of that rise or fall; the antecedent moisture conditions of the watershed; the antecedent geomorphic conditions of the watershed (i.e., the availability and accessibility of glacial sediment [Ahn et al. 2017; Siemion et al. 2023; Wang, Davis et al. 2021; Wang, Gelda, et al. 2021]); the time that has passed since a potentially geomorphically impactful flow; seasonality; and long-term turbidity trends. Hydrological predictors were calculated for Coldbrook and for each of the upstream tributary stations (Table 1). The LGBM-Coldbrook model utilizes only parameters derived from measurements at the Coldbrook gage; the LGBM-Network utilizes parameters from Coldbrook plus all upstream gages (Figure 2). Modeled turbidity values at Coldbrook are thus driven by discharge (and discharge-derived) parameters at only the selfsame gage (LGBM-Coldbrook) or those parameters at both Coldbrook and the upstream stations (LGBM-Network).

3.4 | Model Building and Hyperparameter Tuning

For both LightGBM models, the 7-year observational record was split into training/validation and testing sets of approximately 85% and 15% of the total data. Training and validation data (water years 2017–2022) were used for feature selection and hyperparameter tuning; testing data (water year 2023) were reserved for the final evaluation of model predictive performance (both benchmark and forecasting). Validation data splits were created using an expanding window approach (Hyndman and Athanasopoulos 2018; Mancuso et al. 2021), whereby three different training/validation splits were generated: (1) training 2017–2019, validation 2020; (2) training 2017–2020, validation 2021; (3) training 2017–2021; validation 2022. Dividing in this manner allows for a cross-validation (CV) approach that preserves the time series structure of the data (Goodfellow et al. 2016; Hyndman and Athanasopoulos 2018).

Using each stage of training/validation split, backwards feature selection was performed by running the LightGBM algorithm for each CV split, calculating the mean gain (i.e., the improvement in accuracy that results from the inclusion of a given feature in the decision trees) for each feature across all CV runs, removing the variable that had the lowest mean gain, and re-repeating the process without that feature. During feature selection, we evaluated model performance using three error metrics: the Nash-Sutcliffe efficiency (NSE) (Nash and Sutcliffe 1970), the Kling-Gupta Efficiency (KGE) (Gupta et al. 2009), and root mean squared error (RMSE). We chose error metrics based on the goals of the study and selected multiple to avoid biasing evaluations towards a single performance indicator, which often include metric-specific shortcomings (e.g., sampling uncertainty in KGE and NSE [Clark et al. 2021]). Both the mean values across the 3 CV runs and their standard deviation were evaluated. To determine the best model, the highest weights were given to maximizing NSE and KGE while minimizing RMSE, with secondary consideration given to minimizing the standard deviation of all three metrics across the different CV runs.

Once the best model configuration was selected, hyperparameters were tuned to minimize RMSE using the same three CV splits employed in feature selection. Tuned hyperparameters were those controlling the complexity of the decision trees (num_leaves and min_data_in_leaf) (Table S2). Error

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metrics were calculated with the assistance of the hydroGOF package (Zambrano-Bigiarini 2020). LightGBM models were run using the lightgbm package (Shi et al. 2023), and models were tuned using the tidymodel family of packages (Kuhn and Wickham 2020) in conjunction with the bonsai package (Falbel et al. 2022).

The Quantile-Coldbrook model was trained and tested on the same overall training/testing split (2017–2022 and 2023, respectively). Quantile regression was performed using the quantreg package in R (Koenker 2023).

3.5 | Forecasting & Model Evaluation

Final model configurations were fed operational discharge forecasts at each lead time from both the NWM and NERFC for water year 2023. In raw format, this yielded 355 different time series of discharge (one for each forecast execution between October 1, 22 and September 20, 23) that are each either 204 h (for NWM forecasts) or 72 h (for NERFC) in length. To produce continuous time series with equal length forecast windows (i.e., look-ahead time), we aggregated each timestep into daily bins and combined each bin from each forecast execution (see Supplementary Text S1.1, Figure S2 for additional details). This process yielded continuous time series of discharge and turbidity for 1–3 day and 1–8 day look-ahead times for NERFC and NWM forecasts, respectively. Because models were trained on log-scale values, predicted values must be retransformed to the original units. To correct for the negative bias inherent in the transformation from log space (Rasmussen et al. 2009), we applied a smearing (D) factor to the retransformed predicted values (Duan 1983; Wang, Gelda, et al. 2021). Briefly, D is calculated by retransforming model residuals (in the training data) to linear space and taking the mean of the transformed residuals. Retransformed predictions are then multiplied by D to arrive at a final model prediction.

Models were then evaluated against both observed values and a benchmark generated by feeding observed discharge to each turbidity model (Figure 2). The former provides information regarding the overall forecast skill; the latter, when combined with the former, gives insight into the source of the error (i.e., the turbidity model vs. the forecasted discharge). Model evaluation was undertaken using primarily KGE, NSE, RMSE, percent bias (%Bias), and the decomposition of KGE into its constituent parts: Pearson correlation coefficient (r); a variability term (α), and a bias term (β) (Table 2) (Gupta et al. 2009; Knoben et al. 2019). Though the goals of this work are not a binary evaluation of model skill (i.e., determining "good" vs. "bad" models), to increase ease of discussion here we define thresholds of minimum satisfactory performance: KGE > -0.41 and NSE > 0 (Knoben et al. 2019) and percent bias $> \pm 55\%$ (Moriasi et al. 2007). We utilized these relatively lenient performance thresholds based on the high temporal resolution of our data (hourly) and the constituent of interest (turbidity/sediment) (Moriasi et al. 2015, 2007). To the extent possible, we avoided direct comparison between NERFC and NWM-forced forecasts and focused our discussion on performance differences of each subset (e.g., LGBM-Network vs. LGBM-Coldbrook forced by the NWM).

In addition to evaluation of the entire modeled time series, an event delineation approach (Wasko and Guo 2022) was also employed to evaluate model forecasts during storm event periods. We utilized a local minima method to delineate events, where neighboring valleys are first identified and then an "event" defined by when discharge returns to within a certain threshold of the start discharge, with a minimum spacing of 12 h between peaks. Once delineated, modeled storm events were evaluated in terms of peak turbidity and turbidity "load." Peak turbidity was simply calculated as the maximum observed (or forecasted) value for a given event. Turbidity load was calculated by multiplying the measured turbidity at each time step by instantaneous discharge and integrating over the time length of each event (e.g., Kemper et al. 2019). Though turbidity loading is not, in the strictest sense, a true mass loading rate, given the strong correlation between turbidity and suspended sediment for the study catchment (Spearman $r_s = 0.86$) (Wang and Steinschneider 2022), turbidity load is conceptually equivalent here to suspended sediment load (Wang, Davis et al. 2021), and prior work in the region supports the concept (Mukundan et al. 2018; Peng et al. 2009). Notably, for forecasted values, turbidity load is subject to both error in the discharge term

TABLE 2 | Performance evaluation metrics, formulas, and thresholds for satisfactory performance.

Metric	Formula	Satisfactory	Reference
RMSE	$\sqrt{\frac{1}{N}\sum_{i=1}^{N}(S_i - O_i)^2}$	—	—
NSE	$1 - \frac{\sum_{i=1}^{N}(S_i - O_i)^2}{\sum_{i=1}^{N}(O_i - \mu_O)^2}$	>0	Nash and Sutcliffe (1970); Knoben et al. (2019)
PBias	$\frac{\sum_{i=1}^{N}(S_i - O_i)}{\sum_{i=1}^{N} O_i} \times 100$	$<\pm 55\%$	Yapo et al. (1996); Moriasi et al. (2007)
KGE	$\mathrm{KGE} = 1 - \mathrm{ED_s}$ $\mathrm{ED_s} = \sqrt{[(r-1)]^2 + [(\alpha-1)]^2 + [(\beta-1)]^2}$	≥ -0.41	Gupta et al. (2009); Knoben et al. (2019)
r	$\frac{\sum_i^N (O_i - \mu_O)(S_i - \mu_s)}{\sqrt{\sum_i^N (O_i - \mu_O)^2}\sqrt{\sum_i^N (S_i - \mu_S)^2}}$	—	Gupta et al. (2009)
α	$\frac{\sigma_S}{\sigma_O}$	—	Gupta et al. (2009)
β	$\frac{\mu_S}{\mu_O}$	—	Gupta et al. (2009)

Note: S, simulated (i.e., modeled) values; O, observed values; μ, mean (observed or modeled) values; σ, standard deviation (observed or modeled); r, Pearson correlation coefficient; α, measure of variability; β, measure of bias. Note that the last three terms are the three components of KGE.

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(from discrepancy of streamflow forecasts) and the turbidity term. Event delineation was performed using observed streamflow at the Coldbrook gage and the hydroEvents package in R (Wasko and Guo 2022).

4 | Results

4.1 | Feature Selection for LGBM Models

For the LGBM-Coldbrook model, feature selection yielded a nine-feature model that best predicted turbidity at the Coldbrook station (Table 3, Figure 3). The most important feature for turbidity estimation was, by a substantial margin, log Q as measured at Coldbrook; the least important was the mean antecedent discharge over the preceding month (*1 month mean log Q*) (Figure 3). The only features that resulted in a > 10% loss in accuracy when excluded from the model were log Q, 6 month peak log Q, and 3 month peak log Q.

For the LGBM-Network model, which considered discharge and discharge-derived predictors as measured at the Coldbrook station and each of the other nine upstream monitoring locations, feature selection yielded an eight-parameter model most capable of estimating turbidity (Table 3; Figure 3). Log Q at Coldbrook was again the most important feature, though its relative importance was comparably less than in the LGBM-Coldbrook model (Figure 3). Notably, discharge in Warner Creek (log Q_{Warner}) and Beaver Kill (log Q_{Beav}) were important driving variables in turbidity prediction at Coldbrook. Log Q, 3 month peak log Q, log Q_{Warner}, and 1 week mean log Q_{Beav} all increased model accuracy by > 10% when included. Feature performance was more evenly distributed across parameters in the LGBM-Network model compared to LGBM-Coldbrook. Only a single variable (log Q_{Beav}) had an associated < 5% gain in accuracy in the former, compared to four variables in the latter, and the most important variable in the former was only ~1.3 times more important than the second most, compared to ~2.9 times in the latter (Figure 2).

4.2 | Benchmark Performance

Benchmarks for all three models (Quantile-Coldbrook, LGBM-Coldbrook, and LGBM-Network) were generated by feeding observed discharge values to each model (Table 3). Benchmark model performance was satisfactory for all metrics (Table 2, Table 4). Both LGBM models were demonstrably superior to the relatively simple Quantile-Coldbrook model, though the Quantile-Coldbrook model did observably outperform both ML models in terms of variability (α). Decomposition of KGE indicates the Quantile-Coldbrook model has superior performance in terms of variability (α) to both LGBM models and in terms of bias (β) to the LGBM-Coldbrook model but struggles relative to both in terms of correlation (r), which explains the relatively poorer NSE and overall KGE scores. Percent (%) Bias, a similar but slightly different measure of model tendency towards holistic over- or underprediction (Table 2) to β, suggests that the Quantile-Coldbrook and LGBM-Network slightly overestimate turbidity, while LGBM-Coldbrook more substantially underestimates (Table 4). For high turbidity values and peaks,

TABLE 3 | Features included in each final LGBM model after variable selection.

Variable Name	Explanation	LGBM-Coldbrook	LGBM-Network
log Q	Hourly discharge at Coldbrook (m^3/s)	X*	x*
log Q_{Warner}	Hourly discharge at Warner Ck (m^3/s)		X*
ΔQ	Change in discharge from previous timestep (m^3/s)	X	x
log Q_{Beav}	Hourly discharge at Beaver Kill (m^3/s)		X*
6 month peak log Q	Peak discharge over the previous 6 months at Coldbrook (m^3/s)	X*	
3 month peak log Q	Peak discharge over the previous 3 months at Coldbrook (m^3/s)	X*	x*
1 month peak log Q	Peak discharge over the previous month at Coldbrook (m^3/s)	X	x
1 week mean log Q	Mean discharge over the previous week at Coldbrook (m^3/s)	X	
1 week mean log Q_{Beav}	Mean discharge over the previous week at Beaver Kill (m^3/s)		X
1 month mean log Q	Mean discharge over the previous month at Coldbrook	X	
6 month Δt_{peak}	Time since the 6-month peak discharge occurred at Coldbrook (hours)	X	
1 year Δt_{peak}	Time since the annual peak discharge occurred at Coldbrook (hours)	X	
1 year. Δt_{peak} at Allabem	Time since the annual peak discharge occurred at Allaben (hours)		X

Note: X represents inclusion in each respective model. Asterisk denotes features that result in a > 10% increase in accuracy when included. Site names correspond to specific U.S. Geological Survey discharge gages: Coldbrook, Esopus Creek at Coldbrook (01362500); Warner Ck, Warner Creek (01362357); Beaver Kill, Beaver Kill (01362487); Allaben, Esopus Creek at Allaben (01362200).

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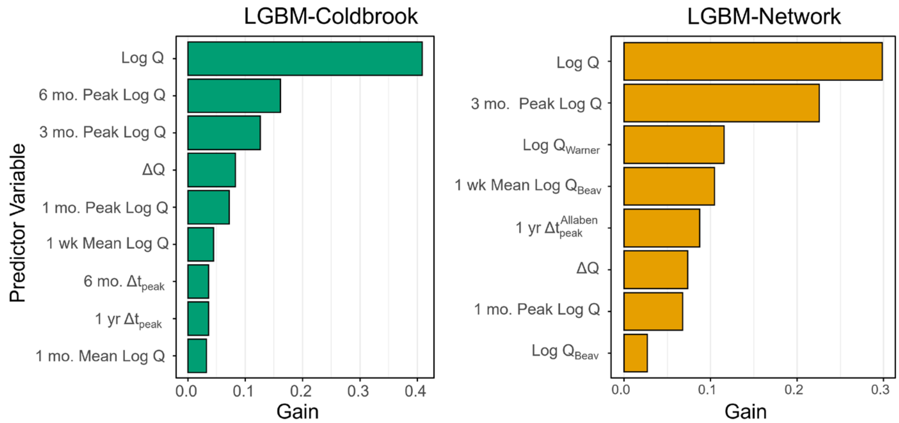

FIGURE 3 | Variable importance for the (a) LGBM-Coldbrook and (b) LGBM-Network model. The x-axis represents the gain in accuracy that resulted from the inclusion of a given variable in the model (i.e., the loss in accuracy when that variable was removed). Units of gain are a fraction (e.g., accuracy decreased by 0.4% or 40% when log Q was removed from the LGBM-Colbdrook model). See Table 3 for definitions of variable names.

TABLE 4 | Benchmark model performance.

Model	RMSE (*FNU*)	NSE	%Bias	KGE	r	α	β
LGBM-Coldbrook	**12.4**	**0.49**	−10.8	**0.59**	**0.7**	0.74	0.89
LGBM-Network	**12.5**	**0.48**	4	**0.61**	**0.7**	0.76	**1.04**
Quantile-Coldbrook	16.1	0.14	**2.7**	0.52	0.53	**0.91**	**1.03**

Note: %Bias, percent bias; r, Pearson correlation coefficient; α, variability term in KGE; β, bias term in KGE. Note that the final three terms (r, α, β) are the three components of KGE (see Table 2). Bolding indicates best performance for a particular performance metric; bolding of more than one row for a given metric indicates that differences in performance are negligible.
Abbreviations: KGE, Kling-Gupta Efficiency; NSE, Nash-Sutcliffe Efficiency; RMSE, root-mean squared error.

the Quantile-Coldbrook model appeared to more frequently underpredict than either LGBM model (Figure 4a), though it did tend to overpredict at low values more consistently as well (Figure 4b). Errors in the Quantile-Coldbrook model were observably heteroskedastic, indicating that prediction with the Quantile-Coldbrook model became increasingly difficult as turbidity values increased (Figure 4a).

Each LGBM model was capable of acceptably modeling turbidity, with an NSE of 0.49 and 0.48 and a KGE of 0.59 and 0.61 for the LGBM-Coldbrook and LGBM-Network models, respectively. Performance between both LGBM models was roughly equivalent across all metrics except for bias, where both metrics (%Bias and β) were observably worse for LGBM-Coldbrook than for Network.

Model capabilities were also evaluated in terms of peak and load prediction for identified storm events (Figure 5; Figure S3). Thirty-four events were identified in water year 2023; no significant differences (p-val > 0.05 using the Kruskal-Wallis test) between modeled and observed storm turbidity load were found (Figure 5a). While not statistically significant, predicted loads using both LGBM models were perceptibly closer to observed event "loads" than those predicted using the quantile model. Models were also fairly accurate when predicting peak turbidity, with a lack of significant difference (p-val > 0.05) between modeled and observed peak turbidity for both LGBM models. Statistically significant differences, however, did exist for event peak turbidity predicted with the Quantile-Coldbrook model (p-val < 0.001) (Figure 5b), suggesting that this model may be less suited for accurate estimations during the most turbid periods of events.

4.3 | Forecast Performance

4.3.1 | General Patterns

We now transition to presenting similar performance statistics as above but for each turbidity model in a forecasting capacity. Models were fed forecasts of discharge (and discharge-derived features, e.g., antecedent mean discharge) from two different forcing products (rather than Q observations)—NWM and NERFC—at different lead times.

In general, turbidity forecasts were most accurate at shorter lead times and became increasingly poor as the forecast horizon increased (Figure 6, Table S3). The overall best performance for forecasts forced by each source, as measured by NSE and RMSE, was for 1-day lead times using LGBM models (Figure 6, Table S3). Given the superior capabilities of flow forecasts when forecast horizons are relatively short (Figures S4 and S6; Table S4), it is relatively unsurprising that turbidity prediction skill follows suit. KGE indicated that the best performance for NERFC-forced forecasts was also at a 1-day lead time, but for NWM-forced forecasts, KGE suggested that performance at longer lead times (3–5 day) was superior to that at shorter.

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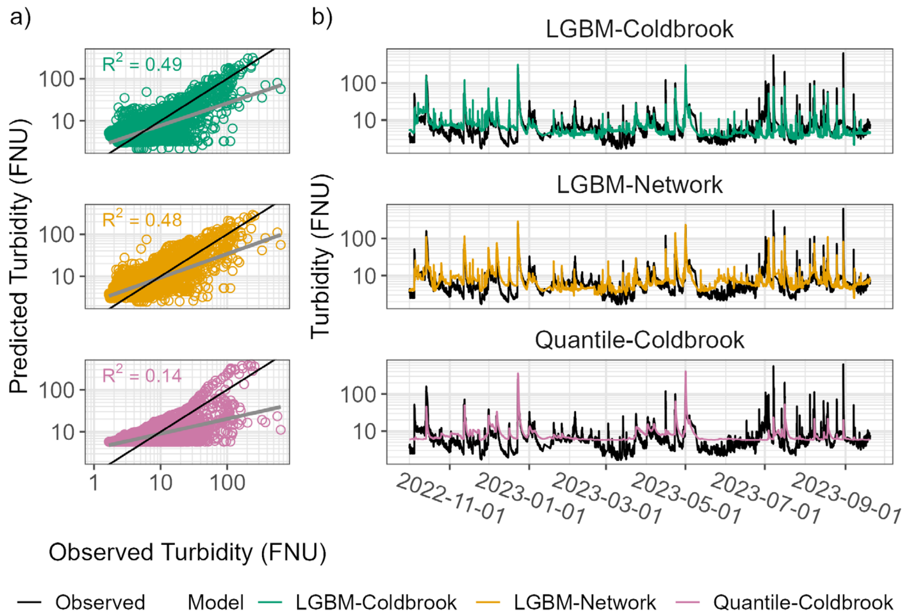

FIGURE 4 | Benchmark model performance. (a) Scatterplots of observed versus predicted turbidity for each model. Solid black line represents the 1:1 line; light gray represents the trend of each. (b) Time series of predicted turbidity (color) and observed (black) for each model.

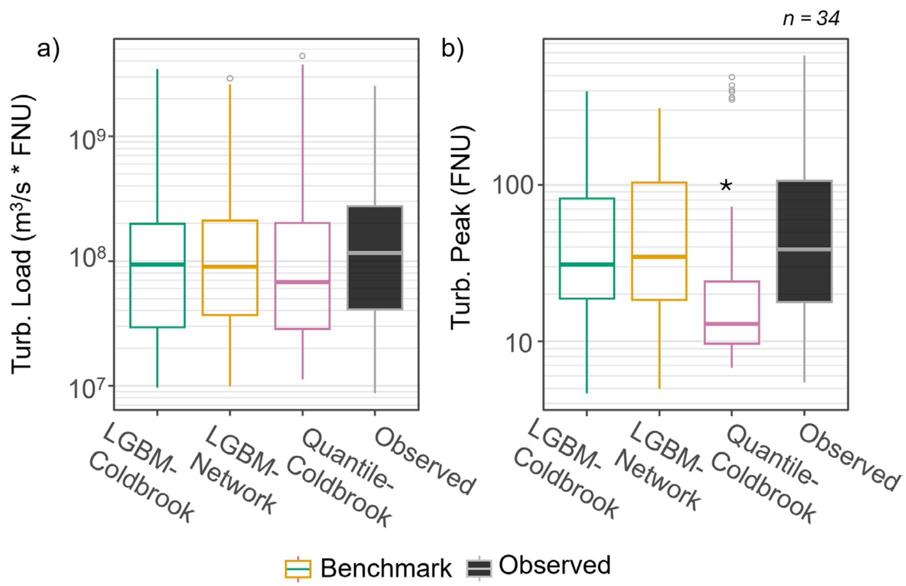

FIGURE 5 | Benchmark versus observed values for (a) event turbidity load and (b) event turbidity peak values for 34 discrete storm events delineated using an event selection algorithm. * denotes significantly different group(s). Open circle indicate outliers.

NSE values declined steadily across the forecast horizon and generally indicate unsatisfactory model performance at lead times beyond 1–3 days, depending on streamflow source. This stands in rather stark contrast to performance as measured by KGE, which consistently indicates satisfactory performance across all forcings and lead times. Decomposing KGE into its constituent parts (correlation, r; variability, α; and bias, β Gupta et al. 2009), we find that β remains relatively constant and fairly satisfactory, varying between ~0.7 and 1.3 (where $\beta=1$ is representative of zero bias), but r and α vary considerably and often depart substantially from ideal values ($r=1$; $\alpha=1$) (Table S3). In concert with NSE and percent bias values, this suggests that models are able to adequately forecast the total "volume" of turbidity (i.e., the overall sum of turbidity values) across the entire forecast horizon but struggle to capture peak values and the shape of turbidity time series as lead times increase.

Except for 1-day forecasts forced by NERFC outputs, all turbidity forecasts consistently underperformed their benchmarks. NERFC-forced forecasts tend toward overprediction (i.e., positive % bias) across the entire forecast domain; conversely, NWM-forced forecasts consistently underpredict (Figure 6, Table S3).

When delineated into event (storm) flow and non-event flow based on observed streamflow at the corresponding timestep, model performance is observably superior during storm events than during non-event periods at lead times up to 6 days (Figure S5). For both LGBM models, this is true across all given combinations of forecast

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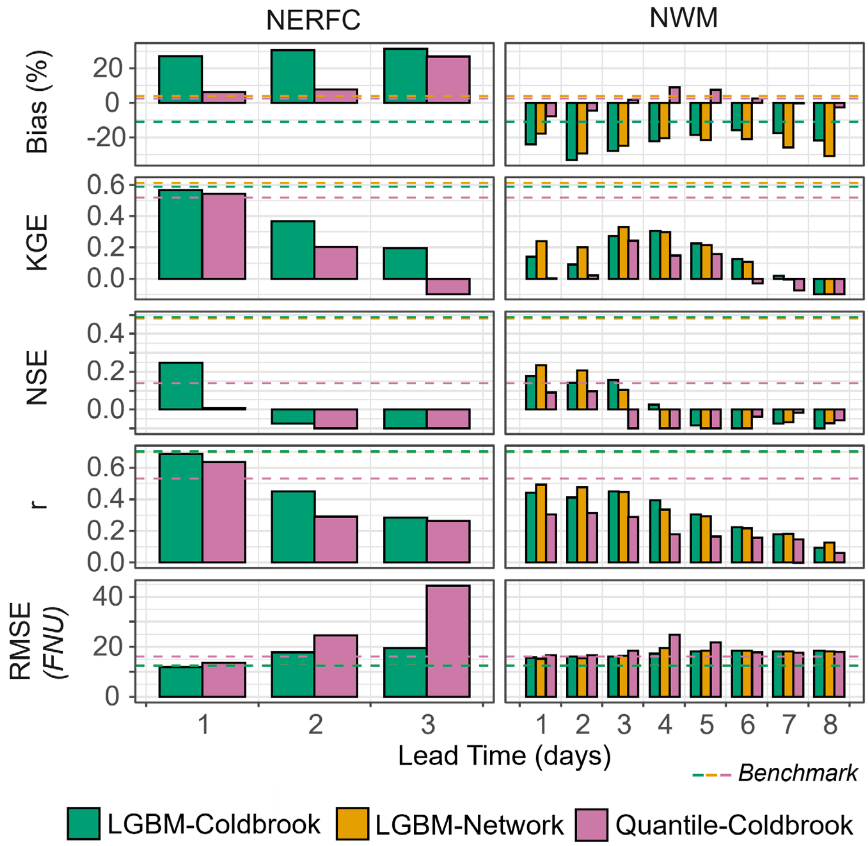

FIGURE 6 | Turbidity forecast (FNU) performance for Esopus Creek at Coldbrook as forced by different streamflow forecast sources. Dotted horizontal lines represent turbidity model benchmark performance, which was generated from feeding observational (i.e., gage) data to models. NERFC, Northeast River Forecasting Center; NWM, National Water Model; Bias—Percent Bias; KGE, Kling-Gupta Efficiency; NSE, Nash-Sutcliffe Efficiency; *r*, Pearson correlation coefficient; RMSE, root mean square error. Values < −0.1 for NSE and KGE have been truncated to improve visualization.

source, lead time, and model. For example, KGE for the LGBM-Network model forced by the NWM with a 3-day lead time is 0.32 for event flow versus 0.02 for non-event flow (Figure S5). Similar to the general patterns seen for the lumped model results, there is a general decline in performance as forecast lead time increases. Though temporal trends in model performance metrics for both lumped and event-delineated results were examined, few patterns or relationships were found. Model performance is markedly poor for non-event flow across all models.

4.3.2 | Inter-Model Differences

For NERFC-derived turbidity forecasts, the LGBM-Coldbrook offers increasing improvements over the Quantile-Coldbrook model with increasing lead times: RMSE decreases by 15%, 28%, and 128% for 1-, 2-, and 3-day lead times, respectively (Figure 6, Table S3). The same pattern is repeated for KGE and NSE, though for this latter metric neither model performs particularly well. Bias for the Quantile-Coldbrook model is noticeably low for all lead times, though this may largely be an artifact of its consistent over-prediction at lower flows and lower turbidities (Figure 4b) rather than a suggestion of overall skill. Turbidity forecasts with the Quantile-Coldbrook model are unsatisfactory beyond a single-day look ahead but arguably acceptable across the three-day forecast horizon with LGBM-Coldbrook (Table S3).

For models forced by NWM streamflow forecasts, improvements over the Quantile-Coldbrook model are offered by both LGBM models. This is generally true for all lead times (Figure 6, Table S3). The LGBM-Network model, which ingests information not only from the Coldbrook gage but from upstream stations, offers an observable improvement over the LGBM-Coldbrook model at shorter forecast horizons (e.g., increase in KGE of ~70%, ~120%, and ~20% for lead times of 1–3 days, respectively) (Table S3). General model performance declines beyond a 4-day lookahead and is arguably unsatisfactory (at least in terms of NSE).

5 | Discussion

5.1 | Improvements Offered by Increasing Model Complexity

Performance metrics for benchmarks and forecasts (i.e., models forced using observed and forecasted hydrologic data, respectively) indicate that LGBM models consistently offered improvements for turbidity prediction over the relatively simple Quantile-Coldbrook model, providing support for our first hypothesis (Table S3, Table 4). In particular, higher values of NSE and *r* (Table 4) in benchmark testing, in concert with the lack of significant differences between observed and modeled turbidity peaks (Figure 5),

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suggest that LGBM models are able to more closely capture the timing and shape of the observed turbidity time series, with only slightly reduced capabilities in matching the magnitude of overall turbidity peaks, agnostic of timing (α in Table 4) (Gupta et al. 2009; Knoben et al. 2019). Observably lower RMSE further suggests that LGBM models are better at avoiding large errors in turbidity prediction than the Quantile-Coldbrook model, though the LGBM-Coldbrook model may tend toward underestimation of turbidity when the entire time series is considered holistically (as indicated by %Bias and the β component of KGE) (Table 4). The enhanced capabilities of these more complex models are consistent with the results of prior work, where time-varying models able to dynamically alter coefficient magnitude(s) have been found to offer relatively superior predictions of turbidity over traditional static relations (e.g., the Quantile-Coldbrook model) (Ahn et al. 2017; Ahn and Steinschneider 2018; Wang, Gelda, et al. 2021). The LGBM models presented here have the additional benefit of being fully unbound by assumptions of linearity (Ke et al. 2017). Together, these results emphasize the potential for LGBM models such as the ones used in this study to offer statistically responsible, skillful water quality predictions.

The benefits of the LGBM models are most pronounced during storm events. Examining the correlation between various parameters as measured at Coldbrook and improvements offered by each individual (benchmark) model in terms of absolute error for a given timestep, it is clear that the improvement offered by LGBM models is strongly positively correlated with the magnitude of instantaneous flow at a given timestep (and the 6-h mean flow) (Figure 7). Similarly, model improvement for each LGBM model over the Quantile-Coldbrook method is most strongly correlated with observed turbidity, suggesting that the more complex machine learning models offer marked improvement over the simplified model during times of high turbidity (i.e., storm events) (Figure 7). Because event turbidity behavior is frequently complex and non-linear both within individual events (e.g., changing concentrations for a given discharge on the rising vs. falling limb) and across seasonal and annual scales (e.g., changes in hysteresis shape and direction) (Asselman 2000; Vercruysse et al. 2017; Wang and Steinschneider 2022), this improvement likely stems from the enhanced ability of machine learning-based models to capture complex sediment transport dynamics and hysteretic behavior (Choubin et al. 2018; Hamshaw et al. 2018; Roushangar et al. 2023). The relatively strong performance of an ML/AI approach thus has particular applicability in highly erosive watersheds where turbidity loads are often notably elevated during storm events, especially when those watersheds drain to reservoirs (e.g., the study basin) (Gelda et al. 2009) or have management concerns otherwise associated with episodic sediment loading during high flows (e.g., post-fire landscapes) (Eidmann et al. 2022). The poor performance of each model at baseflow (Figure S5) suggests that an approach that utilizes an event- and non-event-specific or segmented model (e.g., Underwood et al. 2017) may provide more accurate forecasts across the hydrograph. This is an area for future work.

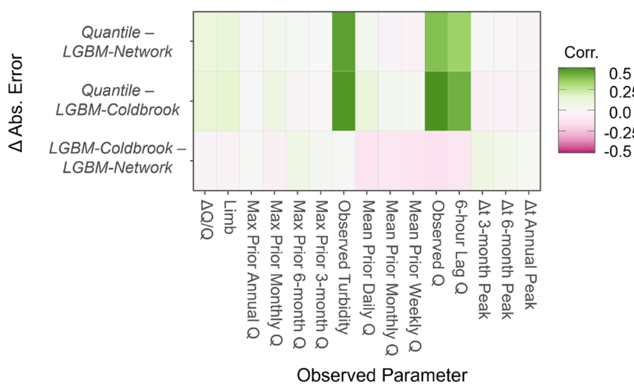

FIGURE 7 | Correlations between various measured parameters at the Esopus Creek at Coldbrook gage (columns) and the differences in absolute error for model pairs (rows). Green colors indicate that the magnitude of the parameter on the x-axis is positively correlated with improvement of the latter listed model over the former (e.g., the larger the observed Q value the greater the error of the Quantile-Coldbrook model relative to the LGBM-Coldbrook model).

In benchmark performance, both LGBM models perform comparably (Table 4), but the difference in bias suggests the LGBM-Network model is slightly better able to capture the turbidity behavior of the entire timeseries (i.e., the area under the time series curve). This can be additionally conceptualized as an enhanced ability of the LGBM-Network model to predict the total "turbidity" volume across the monitored time period (Gupta et al. 2009), which suggests that the LGBM-Network model structure may provide enhanced capabilities for modeling the flux of suspended sediment (as measured by turbidity) into downstream receiving water bodies.

In a forecasting sense, both LGBM models offer improvements in turbidity forecast capability. Forecasts made by the Quantile-Coldbrook model have arguably unacceptable performance beyond 2-day lead times, at least in terms of NSE, which in conjunction with low r values, suggests a relatively poor ability to model turbidity timing and shape (Table 4). This is not entirely surprising, as the Quantile-Coldbrook model is markedly simpler than either LGBM model, both in terms of model architecture and in parameters considered (the Quantile-Coldbrook model, for instance, includes no antecedent terms). It is not our intention to emphasize the relative performance improvement of ML/AI over more traditional linear methods, but to quantitatively highlight the capabilities of LGBM algorithms—which are able to robustly handle large and high-dimensional datasets—using a simplified model (currently utilized by NYCDEP) to add context to performance statistics.

The LGBM-Network model offers performance improvements over the LGBM-Coldbrook model at short forecast lead times (Figure 6, Table S3). This may be due to the fact that the LGBM-Network model incorporates forecasts from multiple locations and is thus more resilient to error in NWM forecasts than the LGBM-Coldbrook model, which relies on the quality of a single forecast at the Coldbrook gage location and more heavily weights (log) Q at the Coldbrook location (Figure 3). At longer lead times, performance between the two becomes increasingly similar (Figure 6), perhaps due to the fact that as the forecast horizon increases and streamflow forecast error becomes more substantial, incorporation of additional forecast locations may only serve to minimally mitigate error. Oddly, KGE suggests performance for NWM-forced forecasts peaks at a 4-day lead time (Table S3), but this is likely primarily an artifact of a spuriously

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accurate forecast for peak discharge on December 23, 2022 (a complex rain-on-snow event) and highlights the need to utilize a suite of error metrics for thorough performance assessment. The enhanced forecasting capabilities of the more complex LGBM models, in conjunction with the relatively better performance of the LGBM-Network model at shorter forecast lead times, provide partial support for our second hypothesis, though the roughly equivalent performance of the LGBM-Coldbrook model in most aspects of benchmark testing suggests that, in this case, improvements were less substantial than we anticipated.

Interestingly, if network-like models—which require the spatially robust forecasts of streamflow only present in the NWM—lead to observable improvements in turbidity forecasts made using data from a singular site, particularly skillful forecasts thus may be achieved by pairing distributed sensor networks with the enhanced spatial resolution present in the NWM. The capability of the LGBM-Network model to achieve similar accuracy in benchmark testing while less heavily weighting the discharge at the gage location for which predictions were made (i.e., Coldbrook) (Figure 3) moreover underscores the intriguing potential of coupling machine learning models trained on distributed sensor networks to spatially dense NWM predictions to offer forecasts more resilient to individual site error. This is potentially relevant for forecasting undertakings in any basin where such networks exist and provides support in favor of their construction in such basins where they are under consideration.

Finally, though benchmark model performance is acceptable, there is ample room for improvement, especially at lower flows. In particular, the models presented here ingest hydrological features alone and would likely benefit from the inclusion of additional data that more directly reflects the geomorphic state of the system (e.g., dynamic connectivity indices [Borselli et al. 2008; Lorenzo et al. 2022]), as well as meteorological data (e.g., Hamshaw et al. 2018). Incorporation of additional (and higher temporal resolution) antecedent turbidity metrics (e.g., antecedent mean daily turbidity) may also improve model performance, although this becomes difficult to consistently incorporate in a forecasting capacity and could serve to decrease performance in longer lead time scenarios where multiple events occur across the forecast window.

5.2 | Water Quality Forecasting Capabilities With Large-Scale Streamflow Models

As is apparent from the previous discussion, post-processing streamflow forecasts from readily available large-scale physical models (such as the NWM) is a viable approach for producing forecasts of water quality parameters in Esopus Creek. Demonstration of the capabilities here emphasizes the viability of a similar approach in basins across the country where sufficient monitoring data exist. Though drivers of turbidity and suspended sediment concentration are complex and non-linear and extend beyond simple flow magnitude (Shin et al. 2023a, 2023b; Vercruysse et al. 2017; Wang and Steinschneider 2022), machine learning models offer the flexibility and robustness to learn from observed data and provide skillful forecasts from hydrologic variables. In Esopus Creek, gradient-boosted decision tree models learned the relatively complex interplay of processes (e.g., dynamic connectivity with lacustrine sediments) that together govern the sediment transport regime of the catchment, suggesting that regime complexity and temporal dynamicity do not preclude skillful prediction. Given the challenges posed by a changing climate and increasing occurrences of extreme events, there is a clear need to expand understanding of the complex feedbacks between altered connectivity and water quality in order to further system knowledge and enhance forecast capabilities of future conditions (Li et al. 2024). To that end, approaches like ours that leverage the flexibility and power of machine learning techniques and employ algorithms (e.g., LightGBM) capable of smoothly handling large datasets with heterogeneous gaps have substantial utility. Examination of the forecasting capabilities of similar models across gradients of hydrologic, geologic, geomorphic, and anthropogenic characteristics is an important area for future work (Li et al. 2024).

From a broad applicability and implementation standpoint, it is encouraging that, outside the 1-day look-ahead window, models forced by the NWM—which has the ability to reach a wide user community due to the high spatial resolution of forecast offerings—had comparable or improved performance to those forced with government-official NERFC streamflow forecasts. Furthermore, turbidity forecasting skill is also likely to improve as streamflow forecasting models improve. Concentrating on the NWM and the capabilities of NWM-forced models to provide accurate forecasts of storm event turbidity "load" modeled error is statistically significantly higher than benchmark error for all models at 1-, 2-, 3-, and 7-day lead times (Figure 8). This suggests there is ample room for improvement on currently offered turbidity forecasts and emphasizes that, as streamflow forecasts improve, turbidity forecasts will more closely match benchmark capabilities (e.g., Figure 5). One potential future avenue towards this enhancement could be the coupling of a turbidity model such as those presented here to an NWM streamflow post-processing model, which has been shown to increase streamflow forecast capabilities (Frame et al. 2021).

Moreover, improvements in streamflow forecasts are likely to continue with each successive model generation. For instance, in the v2.1 iteration of the NWM, which is the generation of the model considered here, over 52% of modeled reaches have a correlation between observed and modeled streamflow greater than 0.8, compared to only 18% of reaches in NWM v1.0 (Cosgrove et al. 2024). Enhanced modeling capabilities are additionally forthcoming as NOAA continues work on the Next Generation (NextGen) iteration of the NWM, which enables model customization based on the most appropriate approaches for the basin or region of interest and facilitates the integration of community-sourced models (Cosgrove et al. 2024). Not only will this undertaking likely improve hydrometeorological forecasts for a particular location, but it will also enable water quality models that ingest NWM outputs to be readily implemented in areas across the country. The approach presented here is to our knowledge the first to utilize the NWM as a water quality forecasting tool and may serve as a guide for others engaged in (or looking to undertake) similar endeavors.

Finally, though the wealth of data and monitoring infrastructure present and available in Esopus Creek is not common, sensor networks continue to expand in catchments worldwide (Hart and Martinez 2006; Kirchner 2006; Kirchner et al. 2004;

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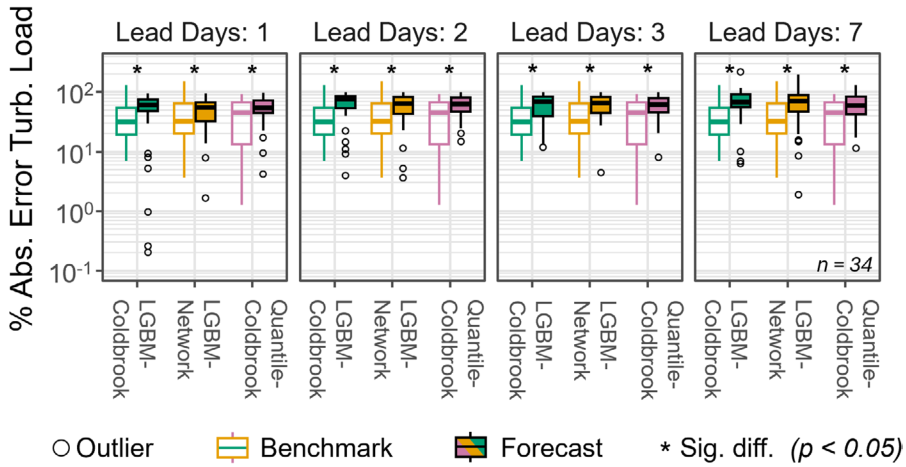

FIGURE 8 | Percent absolute error in storm event turbidity "load" (% abs. error turb. load) for each turbidity model forced by the NWM at lead times of 1, 2, 3, and 7 days for 34 storms in the forecasting period. * denotes benchmark model groupings where the modeled error is significantly different from benchmark. Open circles represent outliers.

Rode et al. 2016). While the methodology offered here is perhaps most readily applied in basins with robust sensor networks and lengthy data sets, the relative skill of the LGBM-Coldbrook benchmark model (0.59 KGE, 0.49 NSE) indicates that capable models of water quality can be built using a single station with a relatively short (5+ year) record. Benefits may be additionally realized if high-frequency data are assimilated into water quality forecast models. Adaptation of our method for basins with relatively sparse or totally absent data records could be an important avenue for future inquiry, as well as the assimilation of observed values. Machine learning models trained on data-rich basins may offer a potentially fruitful way to transfer capabilities to ungauged catchments (Campos and Pedrollo 2021; Kratzert et al. 2019). Overall, the modeling approach and framework delineated here, which utilize publicly available observational and forecasting datasets in conjunction with open-source software, can be readily adapted and applied within any basin where observational datasets are available.

5.3 | Process-Based Insights and Implications for Management

The LGBM-Network model can incorporate network-scale behavior, conditions, and states, and learn spatially heterogeneous relationships between hydrology and turbidity to provide insight into the importance of watershed-scale processes (and the interactions therein) that drive turbidity. Prior work in our study region found that connectivity with glacial legacy sediment, particularly glaciolacustrine deposits with > 90% fine sediment, drives turbidity and SSC levels forced by streamflow (Mukundan, Pierson, Schneiderman, et al. 2013; NYCDEP 2023). Large storm events that expose and erode into such deposits (Figure 9) can drive elevated turbidity levels for a given discharge, which can endure for weeks to months (Siemion et al. 2023; Ahn et al. 2017; Mukundan et al. 2018). Though contact with these basin-wide deposits is distributed heterogeneously across the Esopus Creek basin, field mapping indicates they are disproportionately concentrated in a few tributary watersheds, particularly Stony Clove Creek and Woodland Creek (Figure 1). Warner Creek—whose discharge has substantial importance for the LGBM-Network model (Figure 3)—is the largest tributary to Stony Clove Creek and is a documented source of turbidity within the Esopus Creek catchment (NYCDEP 2023, 2019; Siemion et al. 2023). Additionally, the Beaver Kill tributary—discharge from which is included in the LGBM-Network model—has relatively unique suspended sediment dynamics of the monitored basins: it has both the coarsest suspended sediment as well as bimodal turbidity-discharge dynamics, transitioning from a relatively low turbidity producer/contributor at low flows to a high turbidity producer at high flows (NYCDEP 2023). The importance of Beaver Kill discharge to the LGBM-Network model suggests the bimodal turbidity behavior of Beaver Kill may have a basin scale impact. Overall, inclusion of hydrological parameters for each of these watersheds in the LGBM-Network model emphasizes the influence of reach- and sub-basin scale erosional processes in these tributaries on catchment-scale turbidity behavior.

Both LGBM models also ingest various features reflective of antecedent conditions (peak discharge in the 1, 3, and 6 months prior to a given timestep; the mean discharge in the week and month prior to a given timestep), which are known to commonly influence turbidity and suspended sediment hysteresis patterns (Hamshaw et al. 2018; Vercruysse et al. 2017; Wang and Steinschneider 2022). Interestingly, maximum antecedent discharge at various timescales had a greater importance than mean discharge for predicting turbidity, suggesting that this metric is indicative of a given event's capacity to increase exposure to and contact with glacial lacustrine deposits. This provides further support that (i) turbidity in the Esopus Creek is primarily a function of fluvial contact with and connectivity to these deposits, and (ii) this connectivity scales with storm size, with greater connectivity during and following large events. The inclusion of antecedent maximum discharge at various lookback lengths additionally suggests that the influence of connectivity is complex, with shorter-term behavior interacting with longer-term trajectories. The relatively greater importance of longer-term antecedent discharge (i.e., 3- and 6-month peak discharge have greater importance than 1-month) in both LGBM models indicates the importance of longer-term dynamics (i.e., "memory" of large episodic high flow events) in the system. This is consistent with prior work in the basin, which similarly posits

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FIGURE 9 | Glacial sediment exposed in the valley margins of the Esopus Creek basin. The left shows a mass failure in glacial till (note person for scale). The right shows an exposure of lacustrine sediments in the banks of Woodland Creek. Photo credit: (left) W.D. Davis; (right) J.T. Kemper.

that heightened turbidity for a given discharge occurs due to enhanced connectivity with lacustrine deposits after large storm events and eventually decays back to a more moderate, background regime, potentially until the next large event (Ahn et al. 2017; NYCDEP 2023; Siemion et al. 2023; Wang, Davis et al. 2021; Mukundan, Pierson, Schneiderman, et al. 2013; Mukundan et al. 2015, 2018). The demonstrated capabilities of distributed sensor networks to more fully capture spatially and temporally dynamic connectivity with sediment sources and assist in skillful prediction of sediment and turbidity loading are likely relevant for most watersheds where such heterogeneity exists, especially formerly glaciated or proglacial basins or others where complex distributions of fine-grained sediment deposits (e.g., volcanic ash) are present.

In terms of management for the Esopus Creek catchment, Ashokan Reservoir, and broader NYCWSS, the newly constructed models presented here may be best used in conjunction with existing protocols (NASEM, 2018; Wang, Gelda, et al. 2021) to offer an ensemble forecast of turbidity across a variety of lead times. Utilization of the NWM-forced LGBM-Network model may, for instance, improve anticipation of turbidity loading at moderate lead times beyond a single day (e.g., 2–3 days), where single-station streamflow forecast errors may increase and the incorporation of additional stations may mitigate these errors. Though overall performance metrics may be less than ultimately desirable, watershed managers may be able to use the information provided by the forecasting models here in a holistic assessment of risk for a particular flow event, potentially augmenting existing institutional knowledge and allowing for identification of "high risk" events to be made with greater aptitude. In addition to these immediate opportunities, the models here may be most useful to managers when they are coupled to future versions of the NWM that have improved streamflow accuracy (see Section 5.2). Current models may thus be best employed in a "testing" approach wherein watershed managers run the models with current streamflow forecasts and inspect results to familiarize themselves with the insights and drawbacks offered, but do not base operational decisions on the output. This may enable managers to prepare to incorporate future model iterations into existing protocols. Overall, the capabilities of the models constructed here to further capture non-linear turbidity dynamics can potentially augment existing operational approaches to reservoir management, both in the Ashokan Reservoir as well as in other supply basins in the greater NYCWSS.

6 | Conclusions

Here we have demonstrated the viability of forecasting water quality by coupling data-driven ML models of turbidity to forecasts of streamflow from readily available, process-based hydrological forecasting models (NWM, NERFC). Using the LightGBM implementation of gradient-boosted decision tree models, we leveraged high-frequency turbidity and discharge data from a distributed sensor network and streamflow forecasts from NERFC and the NWM to produce acceptable water quality forecasts at short-to-moderate (1–4 day) lead times relevant to reservoir management entities. A more complex LightGBM model trained on spatially distributed data offered improved predictions for shorter lead times (1–3 days) over a simpler LightGBM model trained on at-a-station only data. We posit that the incorporation of additional sites, made possible by the high spatial resolution of NWM outputs, makes network-scale models (like the LGBM-Network model we present here) more resilient to error in streamflow forecasts. The NWM thus has intriguing potential as a robust water quality forecasting tool capable of offering skillful forecasts at actionable spatial and temporal scales.

ML models were also used to further understand the hydrological processes governing turbidity. Feature weights in both the LGBM-Coldbrook and LGBM-Network models indicate that seasonal-scale antecedent peak discharge conditions exert a substantial control on turbidity at a given discharge. This finding likely reflects the importance of large storm events that increase connectivity with fine-grained glacial legacy deposits via exposure and erosion and the protracted influence of this elevated connectivity through time. Improvements offered by LightGBM turbidity models over a relatively simple regression model suggest that ML models are, to an extent, capable of learning the complex non-linear dynamics that govern turbidity levels and export in the Esopus Creek catchment. Insights and capabilities offered by the models presented here will help to anticipate and manage turbidity loading in the Esopus Creek catchment and the Ashokan Reservoir downstream.

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More broadly, models utilized in this study were constructed using LightGBM, an implementation of gradient-boosted decision trees optimized for performance on large datasets. LightGBM is a readily interpretable, flexible, and computationally efficient algorithm, and thus may represent a robust path forward for harnessing sensor data and streamflow forecasts to make skillful water quality predictions and gain process understanding. To our knowledge, this is the first study utilizing the NWM to produce forecasts of water quality. The approach can be readily implemented or adapted anywhere NWM outputs and monitoring data coincide, and we intend it to serve as a framework for others interested in coupling increasingly available streamflow forecasts to data-driven models to make water quality forecasts for watersheds around the country and globe.

Author Contributions

John T. Kemper: conceptualization, data curation, formal analysis, investigation, methodology, resources, software, validation, visualization, writing – original draft, writing – review and editing. **Kristen L. Underwood:** conceptualization, funding acquisition, investigation, methodology, project administration, supervision, validation, visualization, writing – original draft, writing – review and editing. **Scott D. Hamshaw:** conceptualization, data curation, funding acquisition, methodology, software, supervision, validation, visualization, writing – review and editing. **Dany Davis:** data curation, resources, supervision, writing – original draft, writing – review and editing. **Jason Siemion:** methodology, resources, validation, writing – original draft, writing – review and editing. **James B. Shanley:** conceptualization, funding acquisition, methodology, project administration, supervision, validation, writing – original draft, writing – review and editing. **Andrew W. Schroth:** conceptualization, formal analysis, funding acquisition, investigation, project administration, resources, supervision, validation, visualization, writing – original draft, writing – review and editing.

Acknowledgments

This research was funded by the Cooperative Institute for Research to Operations in Hydrology (CIROH) through a National Oceanic and Atmospheric Administration (NOAA) Cooperative Agreement with the University of Alabama, NA22NWS4320003. The authors are additionally grateful for the expertise, oversight, and advice provided by Rajith Mukundan and Rakesh Gelda, as well as insights and feedback from Scott Lawson and Juli Scamardo. Comments from Jacob Zwart and two anonymous reviewers on an earlier version served to further improve the manuscript. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

NetCDF-formatted output files for each National Water Model (NWM) forecast execution are publicly accessible via the NOAA Operational Model Archive and Distribution System (NOMADS) using ftp and https protocols (for 2 days) or via download from a Google Cloud-hosted archive bucket (https://console.cloud.google.com/storage/browser/national-water-model). Archived Northeast River Forecast Center output files are available for download in ASCII text format via the Iowa State University Iowa Environmental Mesonet (https://mesonet.agron.iastate.edu/nws/). Flow and turbidity data are accessible via the National Water Information System Web Interface (https://waterdata.usgs.gov/nwis). Codes used to access this data and for the other operations detailed in this work (i.e., model training & testing) are available via the CIROH@UVM Github (https://github.com/CIROH-UVM/turbidity-forecasting).

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